

# Annual Report of Year 1

Yi Lu

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# 1 Proposal Aim & Objective

The aim of the proposed project is to exploit the geometry and combinatorics of Newton-Okounkov bodies to deduce positivity properties of divisors on possibly singular projective algebraic varieties, in particular jet ampleness. The aim should be achieved by the following objectives:

- Studying results on and examples of jet ampleness, mainly for toric varieties, but also other classes like abelian varieties, some types of surfaces and Calabi-Yau varieties.
- Where it is possible to calculate Newton-Okounkov bodies, for example on curves and surfaces, explaining the presence or absence of jet ampleness of divisors by the geometry and/or combinatorics of associated Newton-Okounkov bodies.
- Extracting general results, possibly under additional assumptions on the Newton-Okounkov bodies and their underlying valuation semigroups, or by restricting to particular classes of varieties. A promising direction is to study varieties obtained from representation theory like Grassmannians, with their large groups of symmetries making them similar to toric varieties. It is very likely that in contrast to toric varieties with their dense orbit of an algebraic torus action, more general varieties will require looking at whole families of Newton-Okounkov bodies, based on flags roaming the whole variety, to deduce such criteria.
- A general technique to deduce information on general varieties from their Newton-Okounkov bodies is to construct a toric degeneration: Then properties of the toric variety transfer to the general variety to which the toric variety is deformed. Such toric degenerations only exist under special conditions, or their properties must be relaxed. Both types of assumptions may be imposed on the general variety to produce interesting k-ampleness criteria.

## 2 Outline of background and proposed research methodology

Positivity and in particular ampleness of line bundles and divisors is a central tool in algebraic geometry since the beginnings of the subject. Jet ampleness is a relatively new refinement of that notion, usually attributed to Demailly, and heavily studied since the 90s, for example to make progress on Fujita's conjecture [BFS89; BS93]. As always, toric varieties are an interesting testing ground for notions and conjectures on general varieties, so di Rocco studied  $k$ -ampleness on smooth toric varieties in 1999 and found strong criteria relating it to geometry and combinatorics of their moment polytopes [Roc99]. This was generalized to singular toric varieties by Gonzalez and Zhu in 2019 [GZ19].

On the other hand, in the last decade Newton-Okounkov bodies of divisors appeared as a new invariant generalizing moment polytopes of divisors on toric varieties to arbitrary varieties [LM09; KK12; Jow10]. One of the main aims in studying them is to find instances of how their geometry and combinatorics, also of the underlying valuation semigroups, encode more classical properties of polarized varieties, and producing such criteria for jet ampleness will be a milestone in that program. A first step was taken by Kuronya and Lozovanu in 2017 who characterized nefness and ampleness (which can be interpreted as jet ampleness for the most basic jets, the tangent vectors) by the geometry of Newton-Okounkov bodies constructed from specific data [KL17]. One interesting technique to produce such results are toric degenerations pioneered by Harada and Kaveh [HK15].

But it will be already welcome progress if these criteria only work under particular assumptions or for certain classes of varieties. Newton-Okounkov bodies are notoriously difficult to calculate so it makes sense to concentrate first on the some of the examples where they are known, among them surfaces [KLM12], abelian varieties [KL18] and varieties related to representation theory like Grassmannians [RW19]. This suggests the research methodology, starting with examples and working towards generalisations as far reaching as possible. This is an appropriate methodology for a research project of a PhD student, as it allows a gentle start with calculating examples of increasing difficulty, but also leaves open the route to important results worth to be published in top journals.

### 3 Proposed programme of study at CNU and UoL

The research proposal and methodology suggest the following programme of study at CNU and UoL, according to the expertise of the supervisors:

- First one year: Based at CNU, completing knowledge of basic algebraic geometry, toric varieties and Newton-Okounkov bodies.
- One year: Based at UoL, studying  $k$ -ampleness, calculating first examples and proving first criteria. Results found during that period may already be published.
- One year: Based at UoL, exploring generalisations of  $k$ -ampleness characterisations via geometry/ combinatorics of Newton-Okounkov bodies.
- Last one year: Based at CNU, applying final touches and writing up a thesis.

## **4 Any Courses/modules to be taken at CNU and UoL**

The University of Liverpool offers courses on a multitude of mathematical topics delivered through the MAGIC consortium partnering several universities from Northern England to all its PhD students. This allows them to broaden their mathematical knowledge and expertise on areas close to their PhD project and beyond. The exact choice of courses should be tailored to the needs of the student. Furthermore, the Algebraic Geometry group regularly runs reading seminars for its PhD students who for a certain period of time work on and present an algebraic-geometric topic from a monograph or research articles.

## **5 Shared area of research and/or Complementarity of expertise and, or facilities at CNU & UoL**

Zhixian Zhu is an expert in positivity notions on algebraic varieties and its many important conjectures on them. Thomas Eckl studies local positivity and its geometric and combinatorial encoding in convex bodies, in particular Newton-Okounkov bodies, and constructed toric degenerations. Both have several publications on these topics, respectively. Thus, their expertise perfectly combines to supervise the PhD project presented above.

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